

Nonstationarity and change from a statistician's perspective

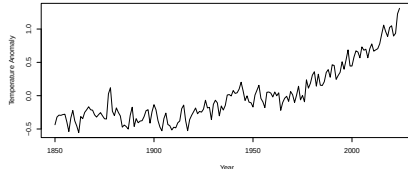
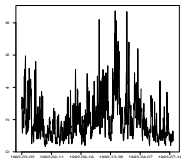
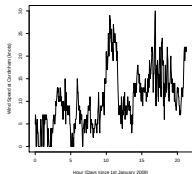
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MMM UCAR Sept 2026



- Definition of nonstationary
- Simplest departure from stationarity
- Taking it further
- Applications interspersed

Statistical properties of a series are **NOT** constant over time.

These arise in a variety of applications including;



Wind Speed

Wave Heights

Global Temperature.

What do I mean?

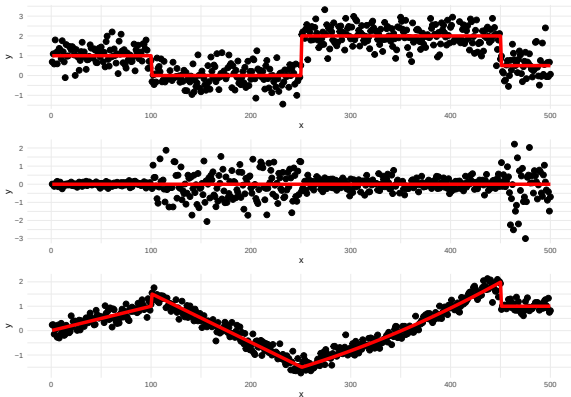
- I mean *NOT* second-order stationary.
- So, mean may *change* with time.
- Unconditional variance may *change* with time.
- Autocovariance, spectrum, etc. may *change* with time.

But how can it change?

What are changepoints?

For data $y_1, \dots, y_n$, if a changepoint exists at τ , then $y_1, \dots, y_\tau$ differ from $y_{\tau+1}, \dots, y_n$ in some way.

There are many different types of change.



What are changepoints?

Changepoints are also known as:

- breakpoints
- segmentation
- structural breaks
- regime switching
- detecting disorder

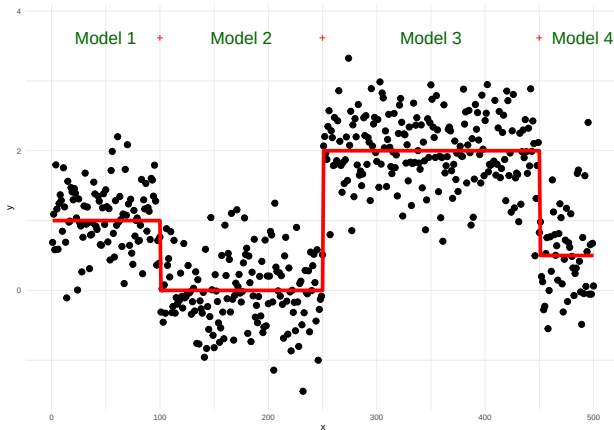
and can be found in a wide range of literature including

- quality control
- finance
- medicine
- environment
- linguistics
- ...

What is the goal?

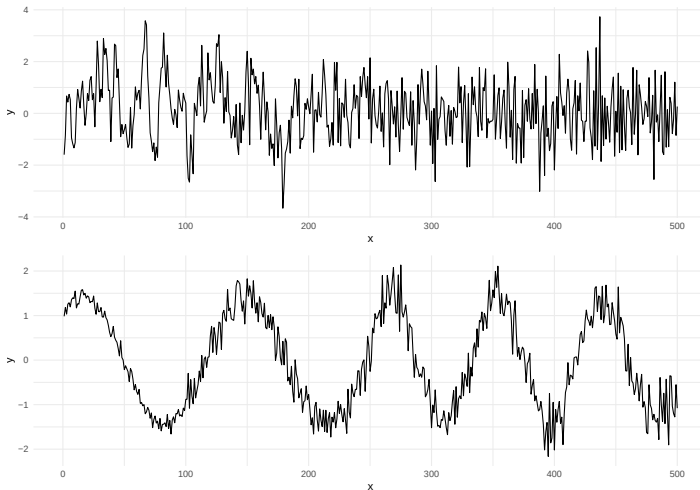
- Has a change occurred?
- If yes, where is the change?
- What is the difference between the pre and post change data?
 - Maybe this is the type of change
 - Maybe it is the parameter values before and after the change
- What is the probability that a change has occurred?
- How certain are we of the changepoint location?
- How many changes have occurred (+ all the above for each change)?
- Why has there been a change?

Problem



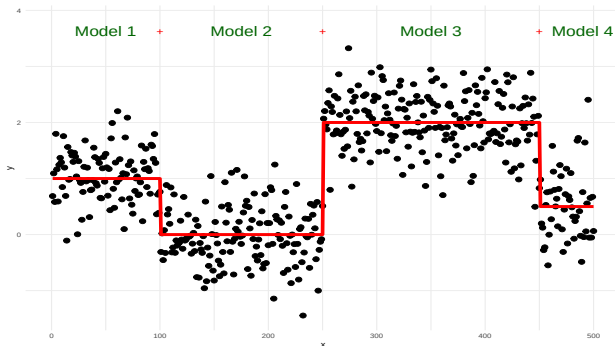
- How many changes?
- Where are the changes? 2^{n-1} possible solutions!

More complicated changes



Interesting Math Challenges

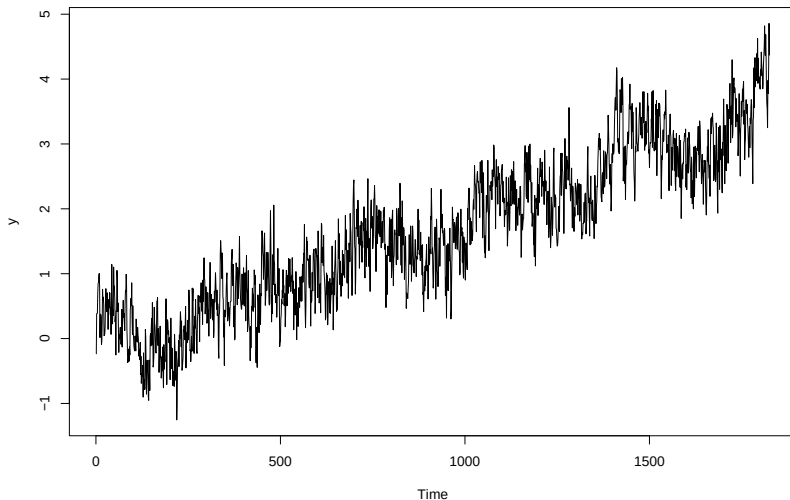
- Optimization
- Model development
- Theoretical properties
- Limits of detection
- Small sample and large sample in the same data



Key practical challenges

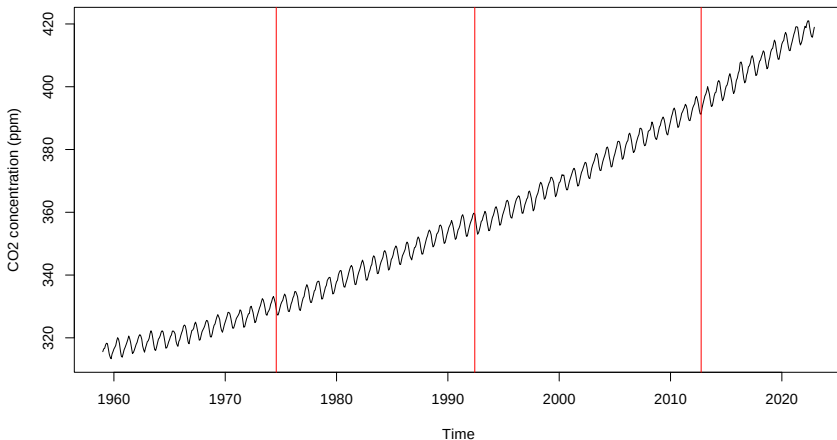
- Model building (assumptions)
- What can change?
- What fitting algorithm to use?
- Penalty selection
- Dealing with outliers

Example data

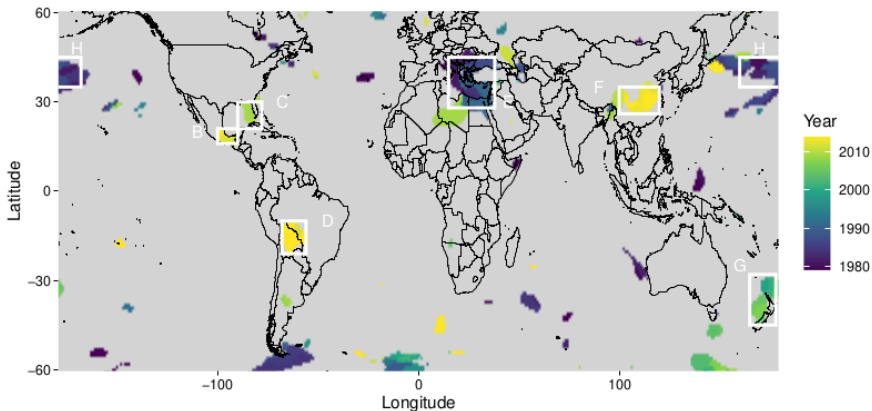


If you wanted to identify changepoints how would you model this?

Example CO2

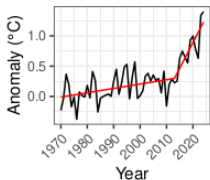


Example Gridded Temp

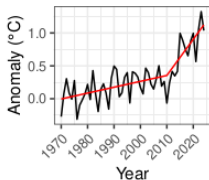


Example Gridded Temp

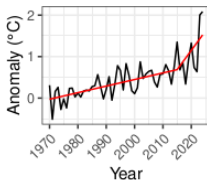
B Southeast Mexico



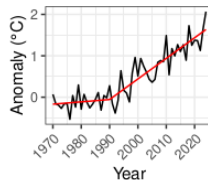
C Gulf of Mexico



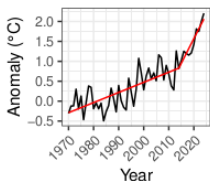
D Bolivia



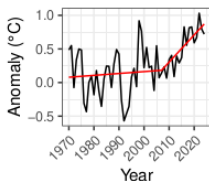
E East Mediterranean



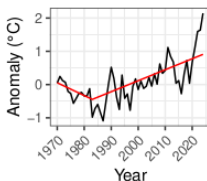
F Southeast China



G New Zealand

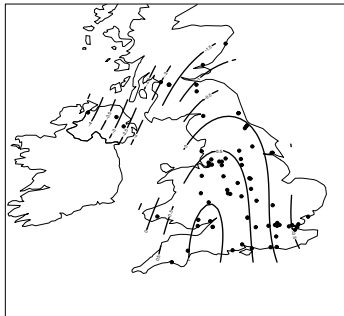


H North Pacific

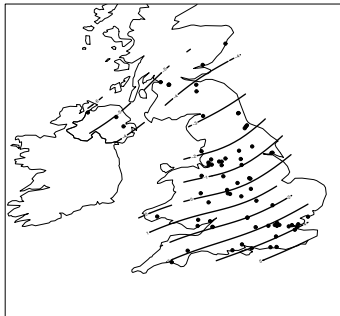


Spatio-temporal NO₂

NO₂ (pre-lockdown)



NO₂ (lockdown)



Spatio-temporal O₃

O₃ (pre-lockdown)

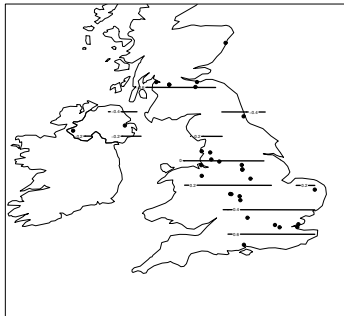


O₃ (lockdown)



Spatio-temporal PM_{2.5}

PM_{2.5} (pre-lockdown)



PM_{2.5} (lockdown)

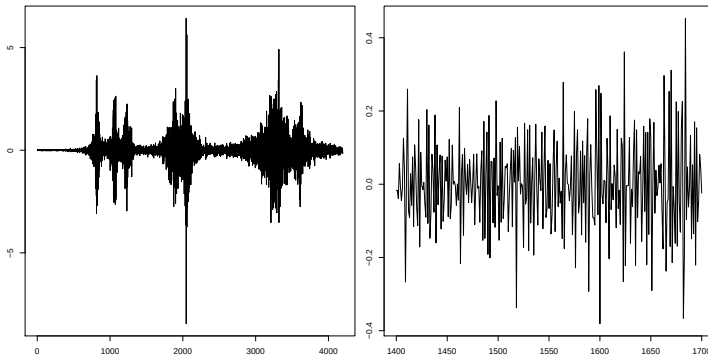


Taking it further

- Regression
 - Trends
 - Seasonality
 - Covariates
- Non-parametric
 - moving average filters
 - splines
 - generalised additive models

Locally stationary

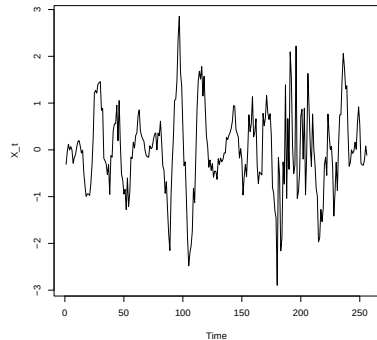
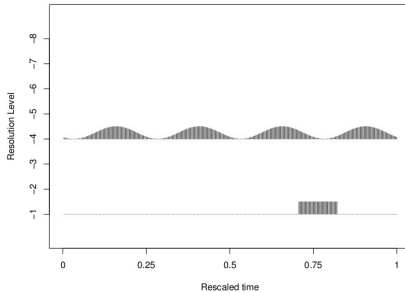
If you take a small enough region, it will appear stationary as the structure varies slowly over time.

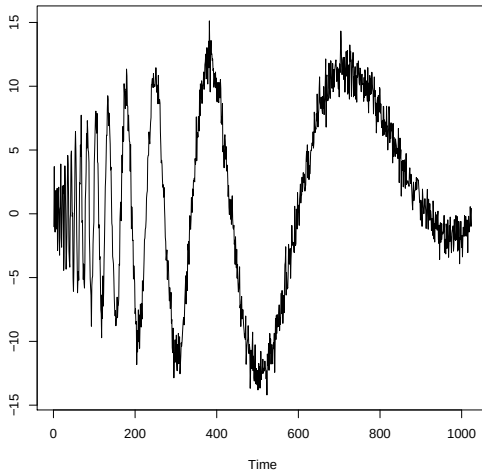


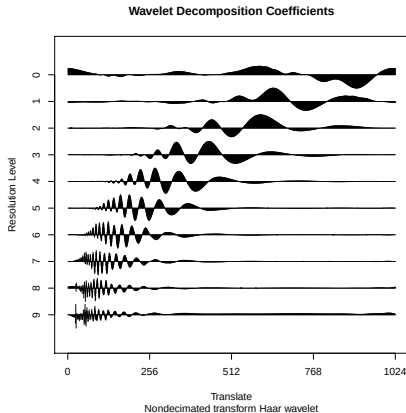
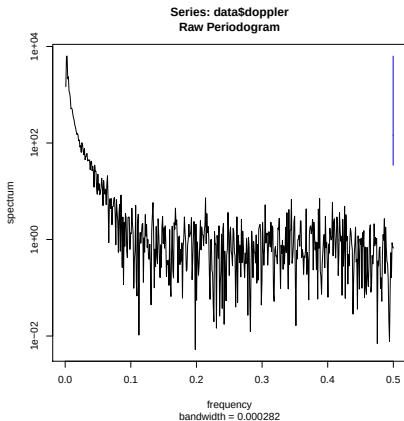
Application areas include;

- medicine, finance, sound
- environmental processes, e.g. wind speeds.

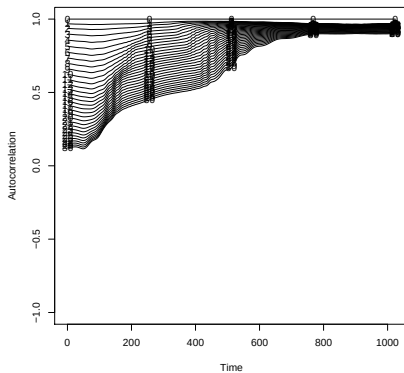
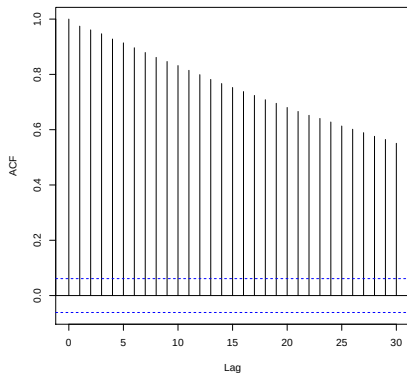
Time varying spectra





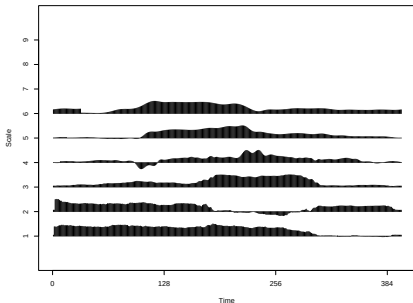


Series data\$doppler

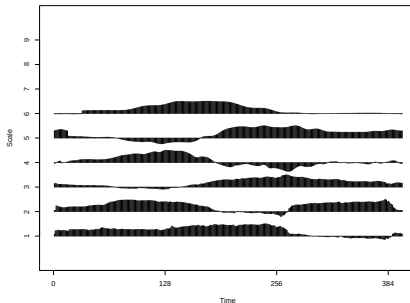


Preliminary Results

Seabreeze4 Spectral Estimate: u



Seabreeze4 Spectral Estimate: v



- Nonstationarities occur in most data
- These can be abrupt (changepoints) or slowly varying (locally stationary)
- Detecting and documenting time-varying properties improves inference
- These techniques are useful in a wide variety of applications
- Talk to me if you want to explore this further!
- I enjoy working with messy real life data . . .
- . . . as often it sparks my next research challenge.